# Lab: Building a Visual Report

Workshop 2 — Module 10

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Data Visualization for Data Scientists
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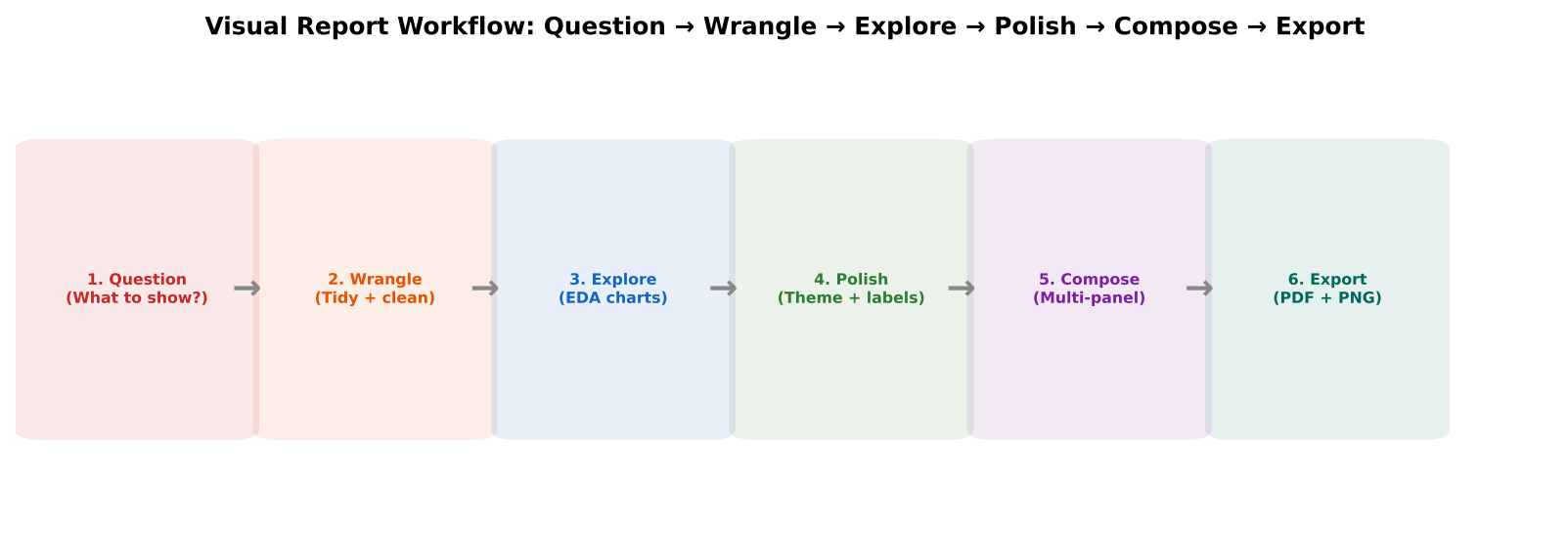
# Module Roadmap

1. The visual report workflow: question to export
2. Apply all Grammar of Graphics layers to six chart types
3. Compose a six-panel dashboard in R (patchwork) and Python (GridSpec)
4. Workshop 2 review: what you now know

**Learning Outcome**
You will produce a complete, publication-quality, six-panel visual report from the Google Play Store dataset, applying every grammar layer covered in Modules 1–9.

# The Visual Report Workflow

# Six Steps: Question to Export

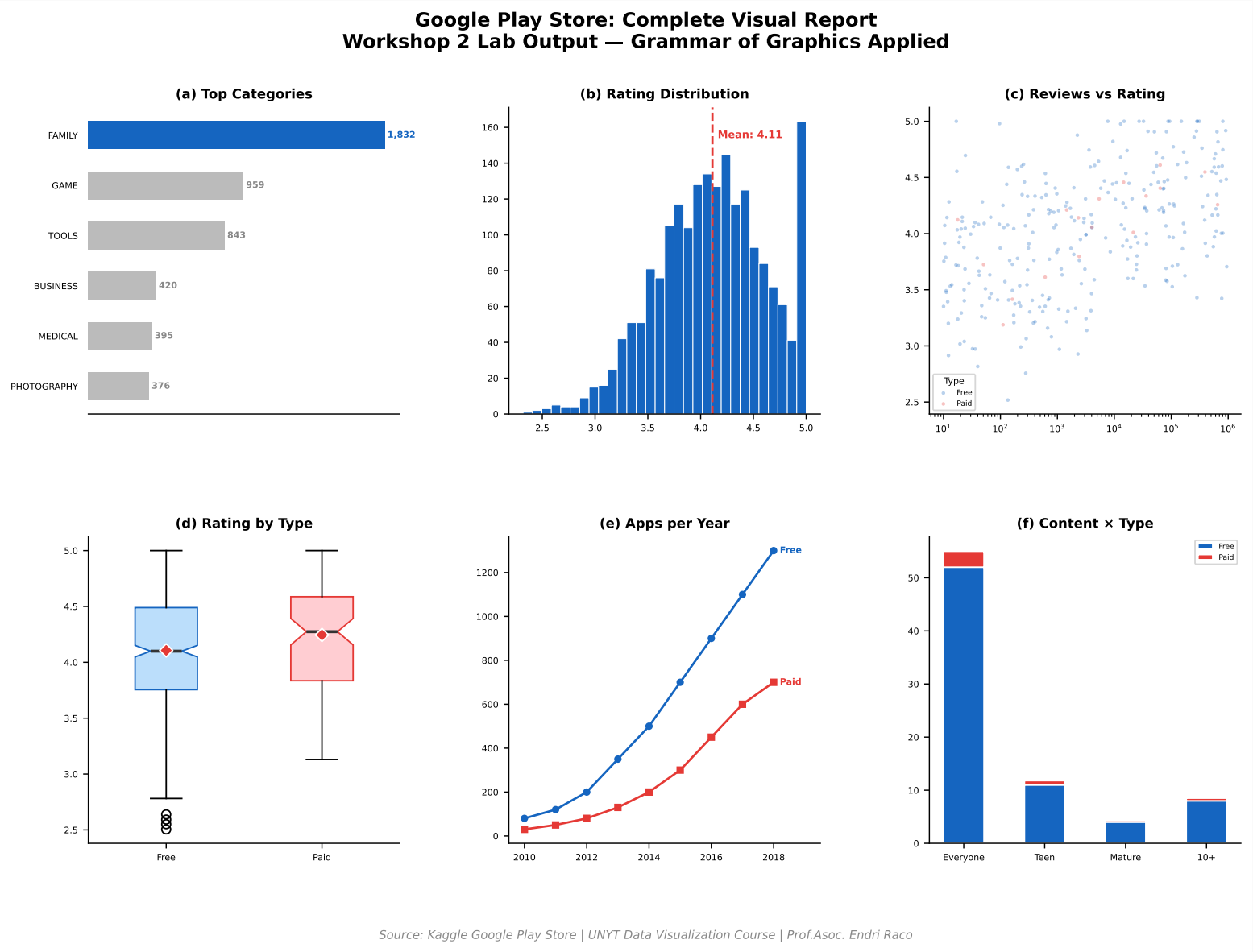



**Pro-Tip**

Start with the question, not the chart type. "Which categories dominate the Play Store?" determines the chart (bar); "How are ratings distributed?" determines the chart (histogram). The grammar layers follow from the question.

# The Target: Six-Panel Dashboard

# Complete Visual Report



# Grammar Layers Applied per Panel

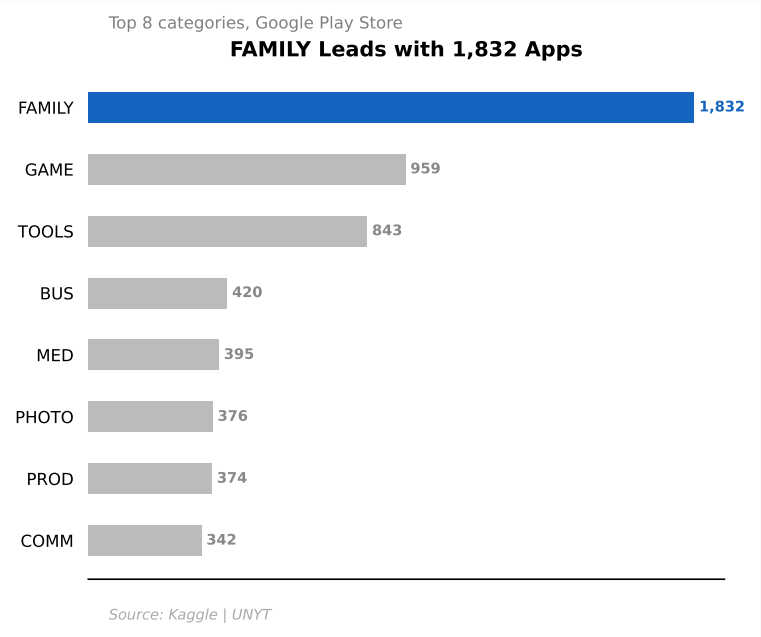
**Grammar Layers Applied per Dashboard Panel**

| Panel | Geom + Coord | Scale + Stat | Theme/Color | Annotation |
|---|---|---|---|---|
| **(a) Bar** | geom_col + coord_flip | fct_reorder, accent | Grey + 1 accent | Declarative title |
| **(b) Histogram** | geom_histogram + geom_vline | stat_bin, bins=30 | Mean ref. line | Distribution shape |
| **(c) Scatter** | geom_point | scale_x_log10, color=Type | hue by Type | Log reveals pattern |
| **(d) Boxplot** | geom_boxplot + stat_summary | notch=TRUE, fill=Type | Mean diamonds | Free ≈ Paid |
| **(e) Line** | geom_line + geom_point | Direct labels, no legend | Direct endpoint labels | Growth trend |
| **(f) Stacked** | geom_bar(position='stack') | fill=Type | Two-colour fill | Composition |

# Building Each Panel

# Panel (a): Top Categories — Bar Chart

```
# Question: Which categories
    dominate?
apps_clean |>
  count(Category, sort = TRUE) |>
  slice_max(n, n = 8) |>
  mutate(
    Category = fct_reorder(Category
        , n),
    highlight = Category == "FAMILY
        ") |>
  ggplot(aes(x = Category, y = n,
             fill = highlight)) +
  geom_col(width = 0.5,
           show.legend = FALSE) +
  geom_text(aes(label = scales::
      comma(n)),
    hjust = -0.15, size = 3,
    fontface = "bold") +
  scale_fill_manual(values =
    c("TRUE" = "#1565C0",
      "FALSE" = "#BBBBBB")) +
  coord_flip() +
  theme_minimal() +
  theme(panel.grid.major.y =
    element_blank()) +
  labs(title = "FAMILY Leads with
```




Grammar layers: `geom_col` + `coord_flip` + `scale_fill_manual` (accent) + `geom_text` (direct labels) + declarative title.

# Panel (a): Same Chart in Python

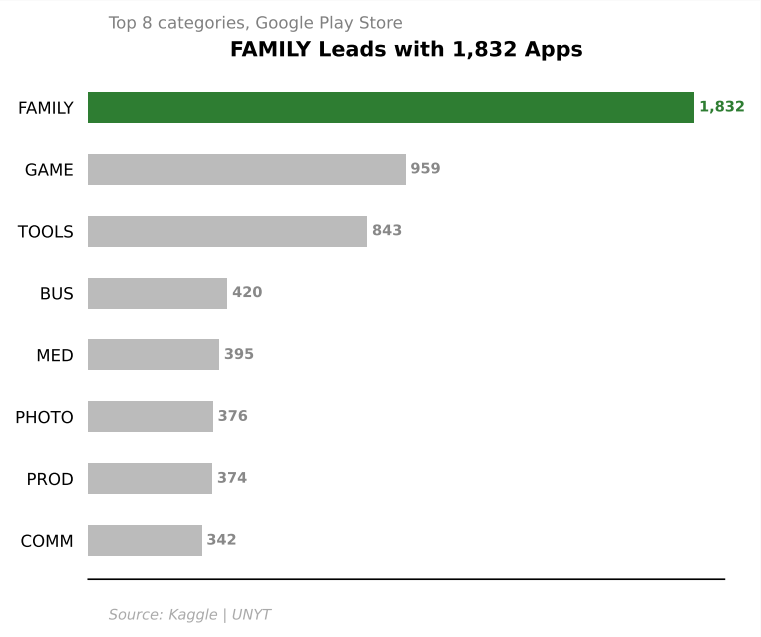
```
top8 = (apps_clean
    .value_counts("Category")
    .head(8)
    .sort_values())

fig, ax = plt.subplots(figsize=(6,
    5))
colors = ["#BBBBBB"] * 8
colors[-1] = "#1565C0"

ax.barh(top8.index, top8.values,
    color=colors, height=0.5)

for i, v in enumerate(top8.values):
    ax.text(v + 15, i, f"{v:,}",
        va="center", fontsize=7,
        fontweight="bold",
        color="#1565C0"
            if i == 7 else "#888")

ax.set_title("FAMILY Leads with
    1,832",
    fontweight="bold")
for sp in ["top","right","left"]:
    ax.spines[sp].set_visible(False
    )
```



# Panels (b)–(f): Grammar Layer Summary

| Panel | R Code Pattern | Python Equivalent |
|---|---|---|
| (b) Histogram | `geom_histogram(bins=30)` + `geom_vline(mean)` | `ax.hist(bins=30)` + `ax.axvline(mean)` |
| (c) Scatter | `geom_point(aes(color=Type))` + `scale_x_log10()` | `ax.scatter(c=)` + `set_xscale("log")` |
| (d) Boxplot | `geom_boxplot(notch=T)` + `stat_summary(mean)` | `ax.boxplot(notch=T)` + `scatter(means)` |
| (e) Line | `geom_line()` + direct labels | `ax.plot()` + `ax.text()` |
| (f) Stacked | `geom_bar(position="stack")` | `ax.bar(bottom=)` |

# Composing the Dashboard

# R: patchwork Composition

```r
library(patchwork)

# Compose 6 panels: 3 on top, 3 on bottom
dashboard <- (p_bar | p_hist | p_scatter) /
             (p_box | p_line | p_stacked) +
  plot_annotation(
    title = "Google Play Store: Visual EDA Report",
    subtitle = "Workshop 2 Lab     Grammar of Graphics Applied",
    caption = "Source: Kaggle | UNYT Data Visualization Course",
    tag_levels = "a"  # auto-labels (a), (b), ...
  )

# Export
ggsave("report.pdf", dashboard, width = 14, height = 9, device =
    cairo_pdf)
ggsave("report.png", dashboard, width = 14, height = 9, dpi = 300)
```

# Python: GridSpec Composition

```python
from matplotlib.gridspec import GridSpec

fig = plt.figure(figsize=(14, 9))
gs = GridSpec(2, 3, figure=fig, hspace=0.4, wspace=0.35)

ax_bar    = fig.add_subplot(gs[0, 0])   # top-left
ax_hist   = fig.add_subplot(gs[0, 1])   # top-center
ax_scat   = fig.add_subplot(gs[0, 2])   # top-right
ax_box    = fig.add_subplot(gs[1, 0])   # bottom-left
ax_line   = fig.add_subplot(gs[1, 1])   # bottom-center
ax_stack  = fig.add_subplot(gs[1, 2])   # bottom-right

# ... plot each panel onto its axes ...

fig.suptitle("Google Play Store: Visual EDA Report",
    fontsize=13, fontweight="bold")
fig.text(0.5, 0.01, "Source: Kaggle | UNYT",
    ha="center", fontsize=8, fontstyle="italic", color="#888")

fig.savefig("report.pdf", bbox_inches="tight")
fig.savefig("report.png", dpi=300, bbox_inches="tight")
```

# Workshop 2 Review

# Workshop 2: What You Now Know

## Workshop 2 Complete: The Grammar of Graphics

| | |
|---|---|
| M01 | **Wilkinson's Grammar**<br>7 layers, algebra, specification |
| M02 | **Wickham's ggplot2**<br>aes, geom, stat, +, template |
| M03 | **Aesthetics & Geometries**<br>5 channels, geom catalogue, overplotting |
| M04 | **Scales & Coordinates**<br>log, date, color, coord, zoom |
| M05 | **Faceting & Small Multiples**<br>facet_wrap, facet_grid, fixed/free |
| M06 | **Themes & Styling**<br>element_*(), custom themes, export |
| M07 | **seaborn & plotnine**<br>3 Python libs, FacetGrid, GoG clone |
| M08 | **Annotations & Storytelling**<br>labs(), callouts, progressive reveal |
| M09 | **Tidy Data & Wrangling**<br>3 rules, pivot, 6 verbs, joins |
| M10 | **Lab: Visual Report**<br>End-to-end, 6-panel dashboard |

# The Grammar Gives You Infinite Charts

| Layer | R | Python |
|---|---|---|
| Data + Aes | `ggplot(df, aes())` | `ggplot(df, aes())` / loop |
| Geometry | `geom_*()` (20+ types) | `ax.*` or `geom_*()` |
| Statistics | `stat_*()` or default | Built into seaborn / manual |
| Scales | `scale_*()` family | `set_xscale` / `cmap` |
| Facets | `facet_wrap/grid` | `FacetGrid` / subplots |
| Coordinates | `coord_*()` family | projection / set_aspect |
| Theme | `theme_*() + theme()` | `rcParams` / manual |
| Labels | `labs()` | `set_title/xlabel/text` |
| Annotations | `annotate() + geom_*` | `ax.annotate/axhline` |

# Key Takeaways

1. A visual report is built in six steps: **question → wrangle → explore → polish → compose → export**.
2. Every chart is a stack of **grammar layers**: data, aes, geom, stat, scale, facet, coord, theme, labs, annotations.
3. **patchwork** (R) and **GridSpec** (Python) compose individual charts into multi-panel dashboards.
4. The **Grammar of Graphics** is a universal system: learn it once in ggplot2, apply it in plotnine, seaborn, or matplotlib.
5. **Workshop 3** builds on this foundation with the Chart Types Encyclopedia: distributions, comparisons, relationships, proportions, rankings, and uncertainty.

# Workshop 2 Homework Summary

**Due: Before Workshop 3**
Create a "Grammar of Graphics Cheat Sheet": implement 15 chart types using GoG layers in both R and Python.

- 15 chart types (bar, histogram, scatter, box, line, area, lollipop, dumbbell, heatmap, violin, density, stacked, faceted scatter, bubble, annotated line)
- Each in R (ggplot2) and Python (plotnine or seaborn)
- One-page PDF per chart showing code + output

Total weight: 5% of course grade.

**Next**: Workshop 3 — Chart Types Encyclopedia (M01: Distributions)